LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 13 Test (GSEB pattern): Our Environment

Chapter / Topic: Our Environment

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SECTION A

Q1. Which group contains only biodegradable items? **[1]**

Answer: (b) Fruit peels, cake and lime juice

Q2. Which of the following is an artificial ecosystem? **[1]**

Answer: (a) Crop field

Q3. Which of the following forms a food chain? **[1]**

Answer: (c) Grass → goat → human

Q4. Organisms that break down the dead remains of plants and animals are called: **[1]**

Answer: (d) decomposers

Q5. Which of the following are environment-friendly practices? **[1]**

Answer: (d) All of the above

Q6. In the food chain grass → deer → tiger, the tiger is a: **[1]**

Answer: (a) secondary consumer

Q7. In a food chain, the energy passed on to the next trophic level is about: **[1]**

Answer: (b) 10% of the previous level (10 per cent law)

Q8. Which of these is a good way to manage household waste? **[1]**

Answer: (c) Composting kitchen waste

Q9. Depletion of the ozone layer is mainly caused by: **[1]**

Answer: (d) chlorofluorocarbons (CFCs)

SECTION B

Q10. What are decomposers? What would happen if there were no decomposers? **[2]**

Answer: Decomposers are micro-organisms, such as bacteria and fungi, that break down the dead remains and waste products of organisms into simpler substances. Without them, dead bodies and wastes would pile up, and nutrients would not be returned to the soil for plants to use.

- Definition with examples: 1
- Consequence: 1

Q11. What will happen if all the organisms in one trophic level are killed? **[2]**

Answer: The food chain will break. Organisms of the next trophic level will lose their food and may starve or die out, while the population of the level below may increase greatly. This upsets the balance of the ecosystem.

- Next level loses food: 1
- Previous level increases / imbalance: 1

Q12. What is biological magnification? Why is it highest in organisms at the top of a food chain? **[2]**

Answer: The progressive build-up of non-biodegradable chemicals, such as pesticides, at each trophic level of a food chain. Organisms at the top eat many organisms from the levels below, and these chemicals are not broken down, so the concentration keeps increasing up the chain.

- Definition: 1
- Reason: 1

SECTION C

Q13. Study the food chain: Grass → Grasshopper → Frog → Snake → Eagle **[3]**

(i) Producer: grass. Tertiary consumer: snake. [1]

(ii) 10 J (10,000 → 1,000 → 100 → 10 J, by the 10 per cent law) [1]

(iii) The eagle. It is at the top of the food chain, and pesticides build up at each level (biological magnification). [1]

Q14. Study the pond food chain: phytoplankton → zooplankton → small fish → big fish **[3]**

(i) Third trophic level (secondary consumer). [1]

(ii) 50 J (50,000 → 5,000 → 500 → 50 J, by the 10 per cent law). [1]

(iii) Much less energy is available at the higher trophic level, so it can support fewer organisms. [1]

SECTION D

Q15. Some years ago, it was proposed that tea on trains should be served in disposable kulhads (clay cups) instead of plastic cups. It was soon realised… **[4]**

(i) Plastic is non-biodegradable and pollutes the environment. [1]

(ii) Loss of fertile top soil. [1]

(iii) Over plastic: paper is biodegradable (and can be recycled). Over kulhads: making paper cups does not use up fertile top soil. [2]

- Each advantage: 1